Suggested problems: 2-4, 6-10

Problem 1: Suppose X and Y are jointly continuous with joint p.d.f.

$$f(x,y) = \begin{cases} \frac{21}{4}x^2 y & : x^2 \leq y \leq 1, \\ 0 & : \text{else.} \end{cases}$$

(a) Compute the conditional p.d.f. $g(y \mid x)$.

(b) Compute $P(Y \geq 1/4 \mid X = 1/2)$.

(c) Compute $P(Y \geq 3/4 \mid X = 1/2)$.

Problem 2: Let X be the amount of time that a person has to wait for service in a queue. The faster the server works in the queue, the shorter should be the waiting time. Let Y stand for the rate at which the server works, which we will take to be unknown. A common choice of conditional distribution for X given $Y = y$ has conditional p.d.f. for each $y > 0$:

$$g(x \mid y) = \begin{cases} ye^{-xy} & : x \geq 0, \\ 0 & : \text{else.} \end{cases}$$

Suppose that Y has p.d.f. given by

$$f(y) = \begin{cases} e^{-y} & : y > 0, \\ 0 & : \text{else.} \end{cases}$$

Compute the joint p.d.f. $f(x, y)$.